



Construction and
Engineering

Business / Technical Brief

PEOPLESFT TO UNIFIER – APPROVED INVOICES
INTEGRATIONS DESIGN SPECIFICATION DOCUMENT

Prepared For: Northwestern Memorial Healthcare

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June 2025, Version 1.0
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Document Control

Change Record

DATE	AUTHOR	VERSION	CHANGE REFERENCE
06/24/2025	Anu Samantha	1.0	Initial Draft

Reviewers

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1 INTRODUCTION

Integration is the process of moving data from one data source to another on a recurring basis. The intention is to keep the target data source synchronized with relevant data from another data source. One is typically the “system-of-record”. The integrations for Northwestern Memorial Healthcare will be developed and configured using Oracle Project Integration Framework (PIF) referred as PIF henceforth on this document.

1.1 Purposes

The purpose of this document is the following:

- Document Oracle’s technical design and process flow for the interface
- As a deliverable, allow Northwestern Memorial Healthcare to approve and sign-off on the technical design.
- Document and maintain changes to the design over time.
- Provide future administrators and developers technical documentation to support the interface.

1.2 Audience

This document is intended for the following audiences:

- Technical resources supporting infrastructure.
- Application administrators and developers
- Northwestern Memorial Healthcare and Oracle Project Managers
- Solution Architects
- Business

1.3 References

This design specification refers the following information sources:

- Meetings and discussions with Northwestern Memorial Healthcare and Oracle teams.
- Field mapping identified by Northwestern Memorial Healthcare and Oracle teams.
- Future Process model
- Oracle’s Solution Design documents

2 INTEGRATIONS OVERVIEW

This document covers the design details for transferring financial data from PeopleSoft to Primavera Unifier using PIF. PIF leverages web services to interface with PeopleSoft and Primavera Unifier.

The interfaces will be scheduled to run within PIF, and it also provides transaction log to view status of the integration jobs and email notifications for reporting transaction details and failures.

The below table details interface that will be implemented within the scope of integrations from PeopleSoft to Unifier. The term “Inbound” and “Outbound” is with the context to Primavera Unifier.

ID	INTERFACE NAME	INTERFACE TYPE	SOURCE SYSTEM	TARGET SYSTEM
I8080	I8080 - PeopleSoft Approved Invoices to Unifier Invoices	Inbound	PeopleSoft	Unifier
I8101	PeopleSoft Approved Change Order Invoices to Unifier Payment Application Invoices	Inbound	PeopleSoft	Unifier

Figure 1: Interface Details

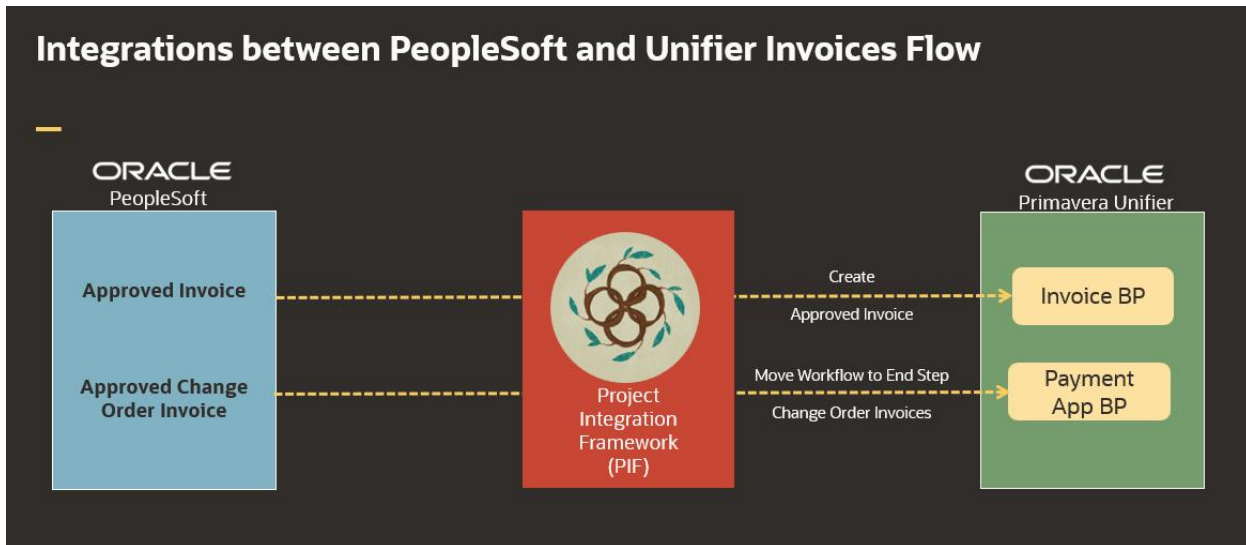


Figure 2: High Level Interface Diagram

3 INTERFACE DESIGN

3.1 I8080- PeopleSoft Approved Invoices to Unifier Invoices (PO)

This interface is used to create Invoices from PeopleSoft View to Invoice BP records in Unifier. Invoices will be updated at header level . Integration will move the workflow to End Step after updating Invoices header details.

3.1.1 Properties of the Data Flow

Interface ID	I8080
Interface Name	I8080 - PeopleSoft Approved Invoices to Unifier Invoices
Source System Name	PeopleSoft
Source System Object Name(s)	NM_UA_INV_023
Target System Name	Unifier
Target System Object Name(s)	Invoice BP
Interface Type	Inbound
Interface Frequency	Nightly- Once
Approximate # of Rows	400 Invoices Per Day
Interface Mechanism	Unifier –Invoice BP PeopleSoft – Views
Table Name Prefix	I8080

Figure 3: I8080 – PeopleSoft Approved Invoices to Unifier Invoices

3.1.2 Business Rules

RULE #	RULE NAME	RULE DESCRIPTION
R-01	Selection Criteria	Read Posted Invoices from PeopleSoft based on Invoice Date
R-02	Invoice Status	Invoices status should be "Posted"
R-03	Invoice Number and Voucher ID	Voucher ID and Accounting DT will be Unique
R-04	PO Number	PO Number should be associated with Invoice.
R-05	Invoice Creation	All Invoices will be created Under project
R-06	Activity Codes	SetID Activity Type Description SHARE01000 Capital SHARE02000 Expense SHARE03000 Initiative
R-07	Cost Codes	Cost Code (Source Type in PeopleSoft) Description 1100 Capital: Architect/Engineer 1200 Capital: Construction Manager 1300 Capital: Consultants 1400 Capital: Regulatory Agency Cst 2000 Capital: Construction 3000 Capital: Insurance 7000 Capital: Capital Contingency 7100 Capital: Capital Tenant Impr 8100 Capital: Medical Equipment 8200 Capital: Furniture/Support Equ 8300 Capital: Technology 9000 Operating Expense
R-08	Line Items/Summary Level	Only Line Items should be brought in for Source Types - 1100, 2000 only. Rest all Source Types should be at header level or Summary level only
R-09	Resource Type	Fetch transactions with Resource_type = ACT from PeopleSoft

3.1.3 Process Flow

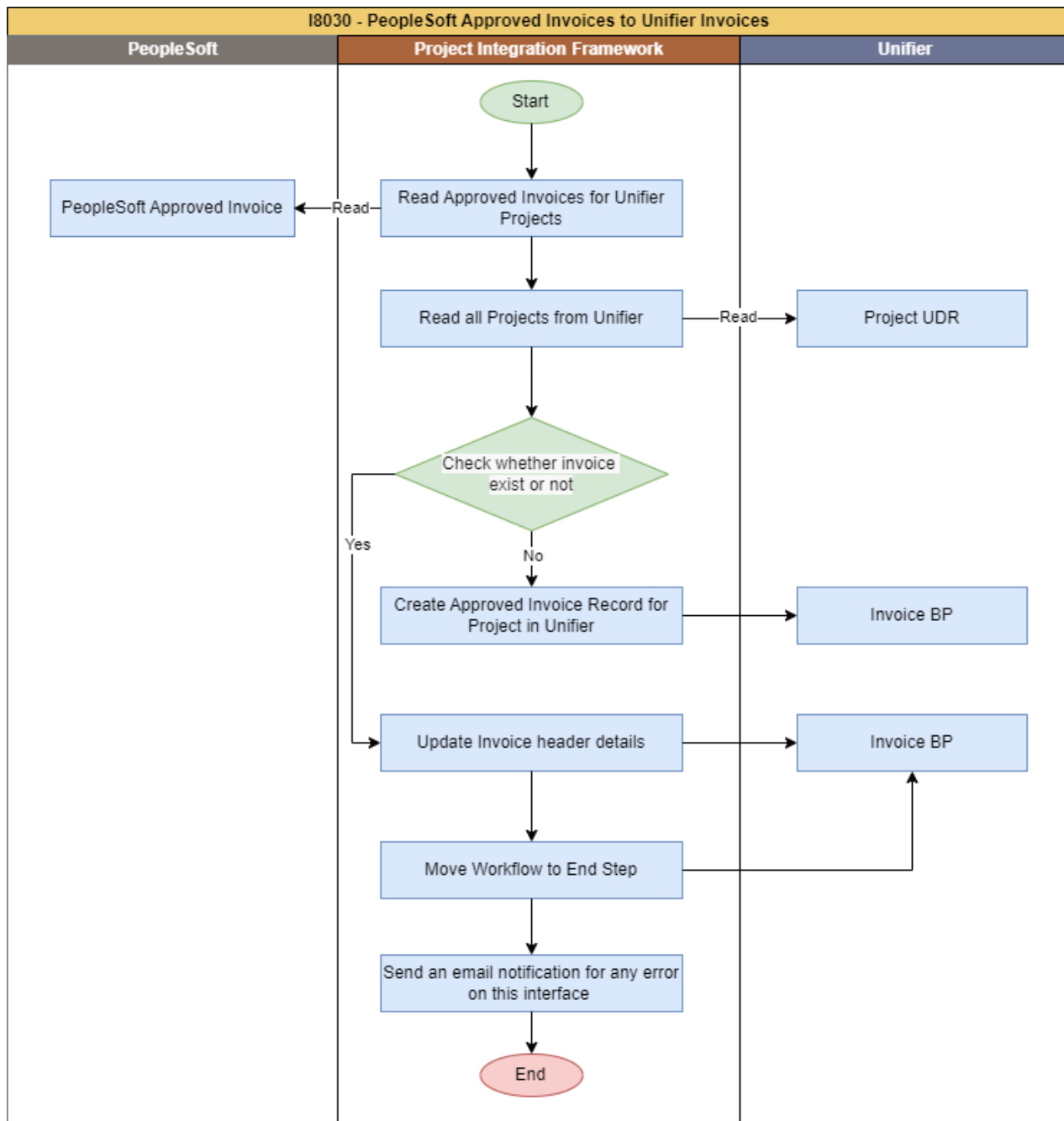


Figure 3: I8080 - PeopleSoft Approved Invoices to Unifier Invoices

1. Clear data in PIF tables (if any) from last interface run.
2. PIF to extract Approved Invoice data for all Projects from PeopleSoft based on PIF last Run Date.

3. PIF will create approved Invoice BP records in Unifier with the Integration Status column set to "New" in PIF staging Table
4. PIF to update Invoice header detail(s) to sync data with recent changes made to PeopleSoft Invoices. This update includes changes to header details.
5. PIF to move the Invoice workflow to End Step after updating header details.
6. PIF will update the field " Integration Status " in the Unifier Approved Invoices BP, setting it to "Sent from PeopleSoft".
7. PIF will generate email error notifications if there is any error while inserting data into Unifier.

3.1.4 Data Mapping

The below is the SharePoint link for data mapping for this interface.

[Oracle-NM-PeopleSoft-Unifier-Invoices-Integration-Mapping.xlsx](#) (Invoice Tab)

3.1.5 PIF Configuration

Interface Name: I8080 - PeopleSoft Approved Invoices to Unifier Invoices Data

To be filled/corrected after completion of interface development.

STEP NO/ INTENT	STEP INFORMATION	BUSINESS RULES / COMMENTS
Step #: 10 Step Name: Truncate PIF tables	Service Name: SQL Service SQL Type: DELETE	Truncate all PIF tables, stage tables
Step #: 20 Step Name: Get Inflight Invoices from PeopleSoft	Service Name: SQL Service SQL Type: SELECT PeopleSoft View Name: NM_UA_INV_023 Table Name: I8080_PSFT_INV	Read Invoices from PeopleSoft
Step #: 30 Step Name: Skip the steps if there is no data from peoplesoft to process	Service Name: GoToService Is Conditional: True Go To Step Sequence: End Process (-1) Table Name: I8080_PSFT_INV_ROWS	Check if there are any Invoices to process or not. If not Skip all the interface steps and move to end step.

STEP NO/ INTENT	STEP INFORMATION	BUSINESS RULES / COMMENTS
Step #: 40 Step Name: Get Projects from Unifier	Service Name: UnifierRESTV2Service Object Name: User Defined Report Method Name: Get UDR UDR Name: I8020_ProjectDetails UDR Level: Project Shell Table Name: I8080_UNI_PROJECTS	Get existing Projects from Unifier using UDR
Step #: 50 Step Name: Check Project Exists in Unifier	Service Name: SQL Service SQL Type: Update Table Name: I8080_PSFT_INV_ROWS	Check whether the given project exists in Unifier.
Step #: 60 Step Name: Get Purchase orders from Unifier	Service Name: UnifierRESTV2Service Object Name: User Defined Report Method Name: Get UDR UDR Name: I8050_PO_REPORT UDR Level: Company Table Name: I8080_UNI_PO	Get existing Purchase Orders from Unifier using UDR
Step #: 70 Step Name: Update pif status if purchase order doesn't exists in unifier	Service Name: SQL Service SQL Type: Update Table Name: I8080_PSFT_INV_ROWS	Check whether the given Purchase Order already exists in Unifier.
Step #: 80 Step Name: Get SOV Line details	Service Name: UnifierRESTV2Service Object Name: User Defined Report Method Name: Get UDR UDR Name: TEST_SOV_LINE UDR Level: Company Table Name: I8080_UNI_SOV	Get SOV Details from Unifier using UDR
Step #: 90 Step Name: Get Invoices from Unifier	Service Name: UnifierRESTV2Service Object Name: User Defined Report Method Name: Get UDR UDR Name: I8080_UNI_INV_REPORT	Get existing Invoices from Unifier using UDR

STEP NO/ INTENT	STEP INFORMATION	BUSINESS RULES / COMMENTS
	UDR Level: Company Table Name: I8080_UNI_INV	
Step #: 100 Step Name: Update pif status if invoice already exists to update the invoice	Service Name: SQL Service SQL Type: Update Table Name: I8080_PSFT_INV_ROWS	Check whether the given Invoices already exists in Unifier.
Step #: 110 Step Name: Update the records for processing	Service Name: SQL Service SQL Type: SELECT Table Name: I8080_PSFT_INV_ROWS	Loop to process one record at a time.
Step #: 120 Step Name: Create Approved Invoice in Unifier	Service Name: UnifierRESTV2Service Object Name: Business Process Method Name: Create BP Record Business Process Name: Invoice Table Name: I8080_PSFT_INV_ROWS	Create new Approved Invoice Record in the Unifier using PIF Staging Table.
Step #: 130 Step Name: Update Actual record	Service Name: UnifierRESTV2Service Object Name: Business Process Method Name: Update BP Record Business Process Name: Invoice Table Name: I8080_PSFT_INV_ROWS	Update Invoice header details in the Unifier as per the PIF Staging Table.
Step #: 140 Step Name: Move workflow to end step	Service Name: UnifierRESTV2Service Object Name: Business Process Method Name: Update BP Record Business Process Name: Invoice Table Name: I8080_PSFT_INV_ROWS	Move all updated Invoices to End Step.
Step #: 150 Step Name: Go to update	Service Name: SQL Service SQL Type: SELECT	Move to the loop step for processing the next record.

STEP NO/ INTENT	STEP INFORMATION	BUSINESS RULES / COMMENTS
step if there are still records to process	Table Name: I8080_PSFT_INV_ROWS	
Step #: 160 Step Name: End Process, Skip to run "Notify Failures" when there are no failed records	Service Name: GoToService Is Conditional: True Go To Step Sequence: End Process (-1) Table Name: I8080_PSFT_INV_ROWS	Check if there are any failed Invoices or not. If not Skip all the interface steps and move to end step.
Step #: 170 Step Name: Notify Invoice failures	Service Name: Email Service Conditional Criteria: Attachment Has Records Attachment File Name: Error Notification : I8080_Invoice Table Name: I8080_PSFT_INV_ROWS	Send an attachment listing all rows which are failed

3.1.6 Error/Warning Handling

ERROR #	RULE NAME	RULE DESCRIPTION
E-01	Approved Invoice Record creation failed	Email will be sent to admin for any failure

3.2 I8101- PeopleSoft Approved Change Order Invoices to Unifier Payment Application Invoices

This interface is used to update Voucher ID on Pencil Draw step and move Workflow to end step when a change Order Invoice is approved from PeopleSoft View to Payment App BP records in Unifier.

3.2.1 Properties of the Data Flow

Interface ID	I8101
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Interface Name	I8101 - PeopleSoft Approved Change Order Invoices to Unifier Payment Application Invoices
Source System Name	PeopleSoft
Source System Object Name(s)	TBD
Target System Name	Unifier
Target System Object Name(s)	Payment Application BP
Interface Type	Inbound
Interface Frequency	Nightly- Once
Approximate # of Rows	50 invoices per day
Interface Mechanism	Unifier –Payment Application BP PeopleSoft – Views
Table Name Prefix	I8101

Figure 4: I8101 – PeopleSoft Approved Change Order Invoices to Unifier Payment Application Invoices

3.2.2 Business Rules

RULE #	RULE NAME	RULE DESCRIPTION
R-01	Workflow End Step	Move Payment Application to End step based on Record Number
R-02	Invoice Status	Invoices status should be "Posted"
R-03	Unifier Project Number and Record Number	Unifier Project Number and Record Number will be Unique.
R-04	Pencil Draw Step	Update Voucher ID and Accounting date from Peoplesoft to Unifier when the record is in Pencil Draw Step
R-05	Final_Voucher_Build Status	Update header details from Peoplesoft to Unifier when the record is in Final_Voucher_Build status.

3.2.3 Process Flow

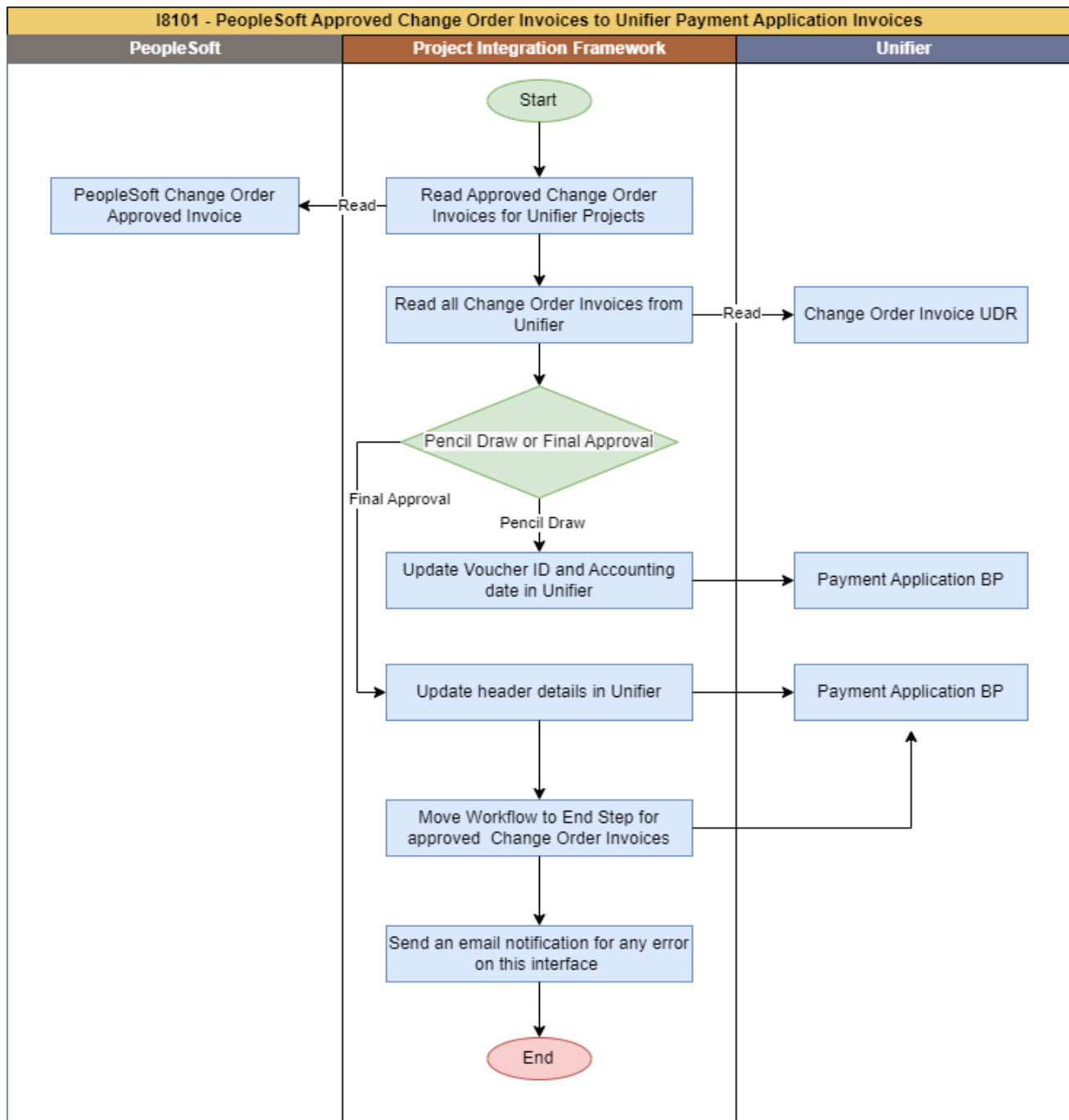


Figure 3: I8101 - PeopleSoft Approved Change Order Invoices to Unifier Payment Application Invoices

1. Clear data in PIF tables (if any) from last interface run.
2. PIF to extract Approved change order Invoice data for all Projects from PeopleSoft based on PIF last Run Date.
3. PIF to update Voucher ID and Accounting date in Unifier when the record is in Pencil Draw status. When voucher is first pushed "PENCIL_DRAW" status.

4. PIF will update the payment application header details when the record is in Final_Voucher_Build status. This includes updating the header details if the same voucher is pushed again in the Final_Voucher_Build status.
5. PIF will move the workflow to the final step for approved Change Order Invoice BP records in Unifier after updating the header details.
6. PIF will update the field " Integration Status " in the Unifier Approved Invoices BP, setting it to "Sent from PeopleSoft".
7. PIF will generate email error notifications if there is any error while inserting data into Unifier.

3.2.4 Data Mapping

The below is the SharePoint link for data mapping for this interface.

[Oracle-NM-PeopleSoft-Unifier-Invoices-Integration-Mapping.xlsx](#) (PayApp Invoice Tab)

3.2.5 PIF Configuration

Interface Name: I8101 - PeopleSoft Approved Change Order Invoices to Unifier Payment Application Invoices

To be filled/corrected after completion of interface development.

STEP NO/ INTENT	STEP INFORMATION	BUSINESS RULES / COMMENTS
Step #: 10 Step Name: Truncate PIF table	Service Name: SQL Service SQL Type: DELETE	Truncate all PIF tables, stage tables
Step #: 20 Step Name: Read Payment Application from PeopleSoft	Service Name: SQL Service SQL Type: SELECT PeopleSoft View Name: NM_UA_INV_034 Table Name: I8101_PS_PAYAPP	Read Payment Application from PeopleSoft
Step #: 30 Step Name: Skip if no data in PeopleSoft	Service Name: GoToService Is Conditional: True Go To Step Sequence: End Process (-1) Table Name: I8101_PS_PAYAPP_ROWS	Check if there are any Payment Application records to process or not. If not Skip all the interface steps and move to end step.

STEP NO/ INTENT	STEP INFORMATION	BUSINESS RULES / COMMENTS
Step #: 40 Step Name: Get Payment Application from Unifier	Service Name: UnifierRESTV2Service Object Name: User Defined Report Method Name: Get UDR UDR Name: I8101_PAYAPPS UDR Level: Company Table Name: I8101_UNI_PAYAPP	Get existing Payment Application from Unifier using UDR
Step #: 50 Step Name: Check Payment Application Exists in Unifier	Service Name: SQL Service SQL Type: Update Table Name: I8101_PS_PAYAPP_ROWS	Check whether the given Payment application exists in Unifier.
Step #: 60 Step Name: Update the pif status to update voucher id and payment details	Service Name: SQL Service SQL Type: Update Table Name: I8101_PS_PAYAPP_ROWS	Update the PIF status according to the record's status in Unifier.
Step #: 70 Step Name: Update the Payment Application records for processing	Service Name: SQL Service SQL Type: SELECT Table Name: I8101_PS_PAYAPP_ROWS	Loop to process one record at a time.
Step #: 80 Step Name: Update the voucher id in the pencil draw status	Service Name: UnifierRESTV2Service Object Name: Business Process Method Name: Update BP Record Business Process Name: Payment Application Table Name: I8101_PS_PAYAPP_ROWS	Update Voucher ID and Accounting date in Unifier for Pencil Draw status.
Step #: 90 Step Name: Update the payment	Service Name: UnifierRESTV2Service Object Name: Business Process Method Name: Update BP Record	Update Payment details in Unifier for Final_Voucher_Build status.

STEP NO/ INTENT	STEP INFORMATION	BUSINESS RULES / COMMENTS
details in the Final_Voucher_Build status	Business Process Name: Payment Application Table Name: I8101_PS_PAYAPP_ROWS	
Step #: 100 Step Name: Move payment application workflow to End Step	Service Name: UnifierRESTV2Service Object Name: Business Process Method Name: Update BP Record Business Process Name: Payment Application Table Name: I8101_PS_PAYAPP_ROWS	Move the workflow to the end step for approved Change Order Invoices in Unifier after updating the header details when the record is in Final_Voucher_Build status.
Step #: 110 Step Name: Go to update step	Service Name: GoToService Is Conditional: True Go To Step Sequence: Update the Payment Application records for processing Table Name: I8101_PS_PAYAPP_ROWS	Move to the loop step for processing the next record.
Step #: 120 Step Name: End Process, Skip to run "Notify Failures" when there are no failed records	Service Name: GoToService Is Conditional: True Go To Step Sequence: End Process (-1) Table Name: I8101_PS_PAYAPP_ROWS	Check if there are any failed Payment Application records or not. If not Skip all the interface steps and move to end step.
Step #: 130 Step Name: Notify Payment Application Failures	Service Name: Email Service Conditional Criteria: Attachment Has Records Attachment File Name: Error Notification : I8101_Payment_Application Table Name: I8101_PS_PAYAPP_ROWS	Send an attachment listing all rows which are failed

3.2.6 Error/Warning Handling

ERROR #	RULE NAME	RULE DESCRIPTION
E-01	Approved change Order Invoice Record workflow update failed	Email will be sent to admin for any failure

4 ASSUMPTIONS

1. All drop down field value should be match between Unifier and PeopleSoft application.
2. The data mapping sheet is designed for the flow of data from one system to another, and if a field is not added, there will be no data flow.
3. PIF jobs are scheduled daily to run automatically, and they can also be run manually on demand by the administrator user.
4. Invoices with Approved/Posted Vouchers will be sent to Unifier from Peoplesoft

5 APPENDIX A: ACRONYMS

ACRONYMS	DESCRIPTION
IB	Inbound
PIF	Project Integration Framework
OB	Outbound
WS	Web Services
RS	REST Services
UDR	User Defined Report

Figure 5: Acronyms

6 APPENDIX B: ISSUE LOG

This Issue Log captures design changes made after sign-off of the use case requirements document.

ISSUE #	DESCRIPTION	STATUS	DATE OPEN	DATE CLOSED	RESOLUTION	COMMENTS

Figure 6: Issue Log